AI Tools for Data Science and Statistics

140.697.79, Summer Institute, 2026–2027

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Instructor

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Course Description

AI tools can accelerate applied work in data science and statistics, but many analysts use them in ad-hoc ways that are inefficient, unreproducible, or unsafe. This course teaches structured, repeatable AI-assisted workflows for coding, data cleaning, debugging, documentation, simulation, and pipeline development. Topics include chatbot and harness selection, context management, skill files, agent workflows, project organization, privacy-aware sandboxing, and cost control. Exercises use R for illustration, but the concepts generalize to any statistical language.

Course Details

Course Number	140.697.79
Term	Summer Institute
Academic Year	2026–2027
Credits	1
Dates	June 15, June 16, and June 17, 2026
Time	9:00–10:50 AM ET
Location	Internet
Instruction Method	Online synchronous
Resources	CoursePlus
Grading	Letter Grade or Pass/Fail
Auditors	Allowed with instructor consent
Undergraduates	Allowed
Prerequisites	No prerequisites
Enrollment	Unrestricted
Instructor Consent	No consent required
Frequency Schedule	One Year Only; only offered in 2026
Format	Lecture, live demos, discussion, hands-on exercises

Course Learning Objectives

By the end of this course, students will be able to:

- Design and implement structured AI-assisted workflows to support coding, data cleaning, debugging, simulation, and documentation.
- Use LLMs and agent-based tools effectively within statistical workflows.
- Critically evaluate AI-generated code, outputs, and recommendations for accuracy, appropriateness, and alignment with analytic goals.
- Apply principles of responsible AI use, including privacy protection, data safety, and ethical considerations when working with sensitive or regulated data.
- Create and manage context files, skill files, and agent configurations to improve model performance.
- Identify and mitigate common pitfalls in AI-assisted statistical computing, such as model inaccuracy, hallucinations, and output bloat.
- Integrate AI tools into epidemiologic and public health data analysis pipelines to improve efficiency, transparency, and reproducibility.

Schedule

Session 1 (Monday, June 15): Introduction, Agents, and Context

We will introduce the course workflow and study the practical landscape of AI tools. The first part of class will provide a condensed introduction to LLMs, why they are useful, where they fail, and why structured workflows matter.

- **Problem set 1 assigned:** AI-assisted data visualization with cleaning, critique, and reflection.

Session 2 (Tuesday, June 16): Models, Agents, and Structured Workflows

We will focus on model selection, agents, and repeatable AI-assisted data science workflows. We will discuss cost, speed, reasoning behavior, coding ability, cloud/local tradeoffs, and when to use different models.

- **Problem set 2 assigned:** AI-assisted statistical modeling with diagnostics and verification.

Session 3 (Wednesday, June 17): Privacy And Simulated Data Workflows

We will study and practice privacy-aware AI-assisted workflows. The class will focus on simulated-data workflows for sensitive or regulated analysis.

- **Final project:** We will start the final project in class; the final project is due July 3, 2026.

Pre-Work

Before the first class, students should complete the Installation Instructions. In brief, students should:

- Install Git or confirm that Git is available on their computer.
- Install R, RStudio or Positron, and Quarto. You may use alternative tools if you are comfortable with them.
- Arrange access to at least one AI tool or agent workflow for the course.
- Clone the course materials and use `git pull` for updates.

A GitHub account is only needed if a student's chosen AI tool requires it, such as GitHub Copilot. Personal access tokens are not part of the course workflow.

Course Project

Each student will complete a simulated-data privacy workflow. The goal is to choose a real dataset, create a simulated version for AI-assisted development, verify the code, then run the reviewed code on the real dataset outside the AI tool. Students may use their own non-sensitive data after discussing it in class. See the final project page for full details.

Date	Milestone
June 15 (Session 1)	Problem Set 1 assigned
June 16 (Session 2)	Problem Set 2 assigned; choose final project real dataset and simulation plan
June 17 (Session 3)	Final project introduced and started in class
July 3	Final project due via CoursePlus

Tools and Subscriptions

Students should arrange access to at least one AI tool or agent workflow for the course. A one-month paid subscription is often useful and typically costs about $20, but existing access or free trials are fine. Options include:

Tool	Cost
OpenAI ChatGPT Plus	$20/month
Anthropic Claude Pro	$20/month
Google Gemini Advanced	$20/month (one month free trial)
Cursor Pro	$20/month
GitHub Copilot Pro	$10/month

Variety across the class is encouraged; we will compare how different tools handle the same problems. If financing is a concern, please reach out to the instructor.

Students should also ensure that Git is installed before the first class. Mac and Linux typically have Git pre-installed. Otherwise, follow directions at git-scm.com.

Course Materials

There is no required textbook. Course readings and further reading are listed on the course website. Course materials will be distributed through CoursePlus and the public course repository.

Methods of Assessment

This course is evaluated as follows:

- **Participation (30%):** Attend all sessions and engage actively in discussions and hands-on exercises.
- **Problem sets (30%):** Complete applied visualization and modeling exercises that practice structured AI-assisted workflows.
- **Final project (40%):** Complete a simulated-data privacy workflow at home with separate `simulated/` and `real/` folders and submit a reflection.

Generative AI Policy

Using AI tools is the subject of this course. Their use is permitted, encouraged, and expected. Students are responsible for understanding, verifying, and explaining any AI-assisted work they submit.

Academic Ethics

Students enrolled in the Bloomberg School of Public Health of The Johns Hopkins University assume an obligation to conduct themselves in a manner appropriate to the University's mission as an institution of higher education. Students should be familiar with the policies and procedures specified under Policy and Procedure Manual Student-01 (Academic Ethics) and the Student Conduct Code (Student-06), available at my.publichealth.jhu.edu.

Student Health and Well-being

If you are struggling with anxiety, stress, depression, or other mental health related concerns, please consider connecting with resources:

- Student support: bit.ly/bsphstudentsupport
- Mental Health Services: wellbeing.jhu.edu/MentalHealthServices
- Behavioral Health Crisis Support Team (24/7): 410-516-9355

Disability Accommodations

Student Disability Services (SDS) provides accessible and inclusive educational experiences for students with disabilities. To request accommodations:

1. Complete the SDS online application via AIM
2. Submit documentation using the provided link after application submission
3. Schedule a meeting with Audrey Ndaba

More information: Student Disability Services